A3-PS020-TC511

PS-020

Local SRv6 Path Steering and Measured Failure Recovery

6G & Future Networks | AIORI Hackathon 3

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IP
ROUTING
×
SRv6

PATH STEERING
FAILURE RECOVERY

Abstract & Problem Description

Problem statement

Compare ordinary hop-by-hop IPv6 routing with explicit SRv6 path steering in a five-router virtual testbed, using the same controlled link failure.

Challenge

IPv6 routing selects a next hop at each router. SRv6 can steer packets through an ordered list of Segment Identifiers (SIDs).

Experiment

Configure two explicit SRv6 paths, introduce one controlled link failure, and activate a backup policy using a script.

Evidence

Use packet captures and repeated trials to measure packet loss, interruption, failure detection, repair delay and SRv6 header overhead.

Planned methodology only: measured values will be added after running the GNS3 testbed.

Background

Ordinary IPv6 routing

Routers consult IPv6 routing tables and forward packets to the next hop. Failover depends on configured routing and failure detection.

Segment Routing over IPv6

SRv6 uses SIDs—IPv6-address-form identifiers representing segments or behaviors—to guide packets through configured waypoints.

Failure recovery

A failed link can interrupt traffic until failure is detected and a route or policy change restores forwarding.

Why measure?

The same topology, endpoints, traffic and failure enable a fair comparison instead of assuming either method is faster.

Keywords: IPv6 · SRv6 · SID · SRH · packet loss · interruption time · failure detection · repair delay

Literature & RFC Review

Primary protocol reference

RFC 8754 — IPv6 Segment Routing Header (SRH)

Defines the SRH format used to carry segment-routing information in IPv6 packets. It is the principal reference for inspecting SRv6 packet headers and SID lists in packet captures.

Standards check

Use the RFC to identify SRH fields and segment-list structure.

Packet evidence

Capture packets and verify expected SID traversal.

Scope

Measure this testbed only; do not claim universal SRv6 superiority.

Reference: https://www.rfc-editor.org/rfc/rfc8754.html

Proposed Solution

01 Build testbed

Five routers; two viable paths; IPv6 connectivity.

02 IPv6 baseline

Configure hop-by-hop IPv6 routing and verify reachability.

03 SRv6 paths

Configure SIDs and two explicit SID lists on a supported platform.

04 Link failure

Shut the selected link and run the backup-policy switch script.

05 Measure

Capture packets; calculate loss and recovery timing.

06 Repeat

Run identical trials and compare observed results.

Fairness controls: same topology, endpoints, packet size/rate, failure link and trial duration.

Solution Optimization

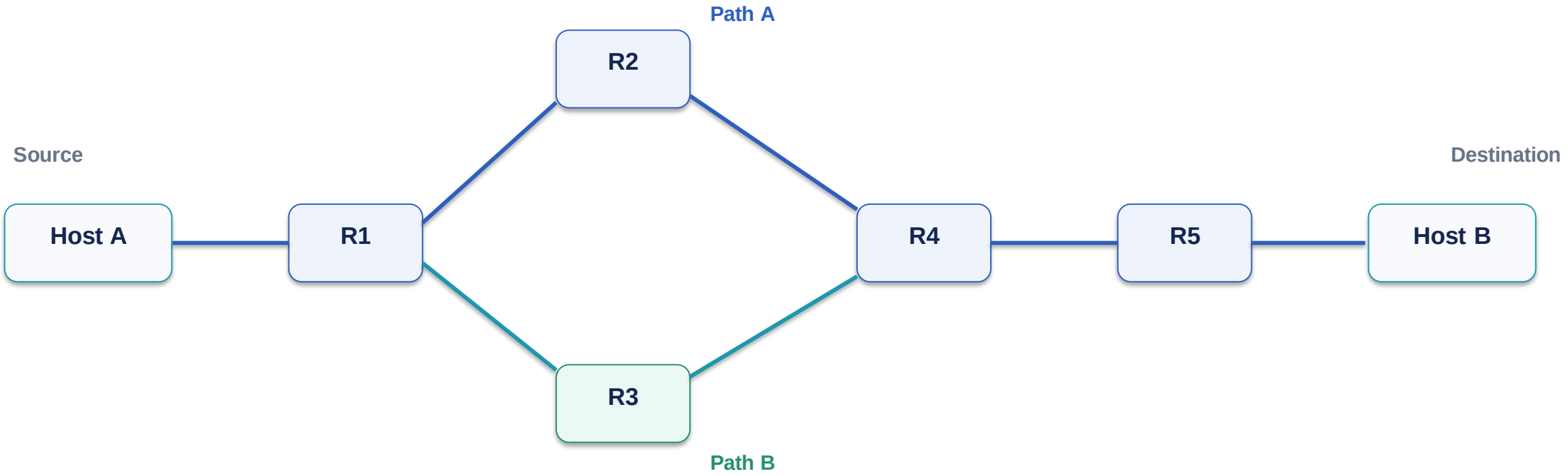
Optimization proposed: improve repeatability, observability and fairness.

Aspect	Initial approach	Improved experiment design
Path selection	IPv6 routing table	Explicit SID list for SRv6 steering
Failure handling	Observe route recovery	Scripted switch to a preconfigured SRv6 backup policy
Traffic measurement	Occasional ping checks	Timestamped probes and packet captures
Path verification	Ping proves reachability only	Inspect SRH and confirm SID traversal
Reliability	Single trial	Repeated trials; report mean/range and setup

No performance winner is assumed; conclusions must follow measurements.

Solution Architecture & Design

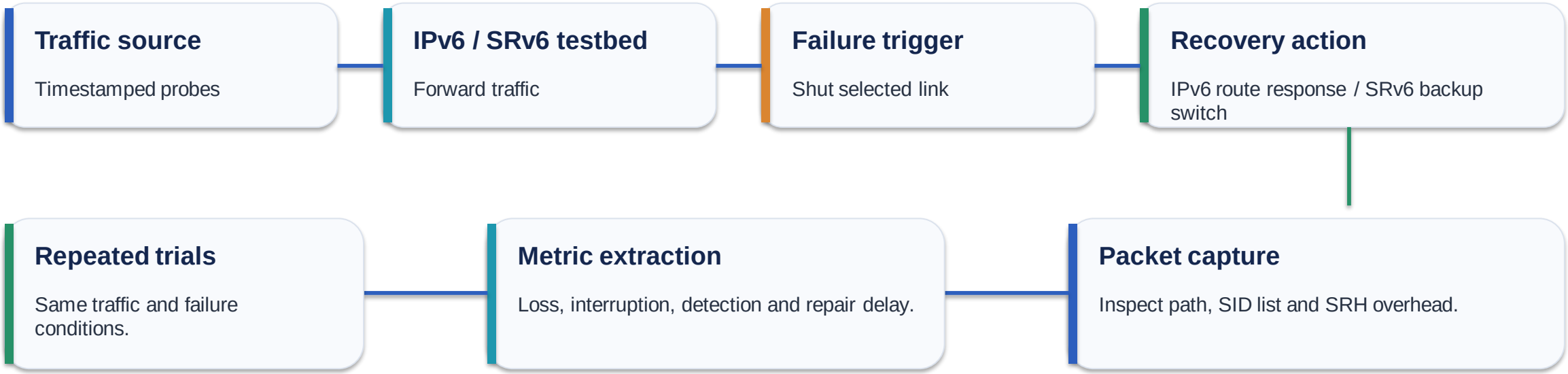
Illustrative five-router testbed — confirm the selected router image supports SRv6.



Ingress policy + IPv6 forwarding + packet capture

Illustration only: interface addressing and SID behaviors depend on the chosen software image.

Data Flow Diagram



Deliverables: PCAP captures · trial logs/CSV · comparison table · technical report · demo

All values will be collected during implementation; no measured results are claimed here.

Deployment Timeline

Proposed implementation milestones

1	PHASE 1	**Platform & topology**	Select compatible image; build five routers; verify IPv6 reachability.
2	PHASE 2	**IPv6 baseline**	Configure routes and record baseline/failure behavior.
3	PHASE 3	**SRv6 setup**	Configure SIDs, two explicit paths and packet-capture verification.
4	PHASE 4	**Failure automation**	Trigger link-down and scripted backup-policy activation.
5	PHASE 5	**Measurement**	Repeat trials; extract loss, interruption, detection/repair delay and overhead.
6	PHASE 6	**Submission**	Finalize report, demo, code and GitHub deliverables.

Align timings with the official AIORI hackathon milestones.

Bibliography

[1] F. Clad et al., “Segment Routing Header (SRH) for IPv6,” RFC 8754, Internet Engineering Task Force, March 2020.

https://www.rfc-editor.org/rfc/rfc8754.html

[2] “PS-020: Local SRv6 Path Steering and Measured Failure Recovery,” AIORI Hackathon 3 problem statement, supplied project brief.

Add further papers only after reviewing and using them in the implementation or report.

THANK YOU

Questions & Discussion

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Thank you for your attention!

SRv6

PATH STEERING
FAILURE RECOVERY
MEASURED RESULTS